



Interpretable Tree Policies Beyond On-Policy RL: Reproducing, Extending, and Stress-Testing SYMPOL

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Overview

SYMPOL (Marton et al., 2024) trains a *differentiable decision tree* as an RL policy, keeping a tree’s interpretability while learning end-to-end — but it was only ever demonstrated with PPO (on-policy). We (i) **reproduce** the SYMPOL baseline suite, then push the interpretable tree into two settings it was *never tested in*: **(E5)** off-policy training with Soft Actor-Critic, and **(E6)** robustness under environment and actuator perturbation. Our findings *extend*, and do not contradict, the original work.

Research Questions

- **Q0 — Reproduction.** Do MLP and Soft-Decision-Tree (SDT) baselines under PPO match the performance envelope reported by SYMPOL across 11 environments?
- **Q1 — Off-policy feasibility (E5).** Can the hard-routed SYMPOL tree be trained inside SAC, an off-policy actor-critic with a reparameterised, entropy-regularised objective?
- **Q2 — Robustness (E6).** Do axis-aligned tree policies degrade *more gracefully* than a dense MLP when the deployment environment drifts?

Background

MLP — fully-connected actor; the opaque-but-strong baseline. **SDT** (Frosst & Hinton, 2017) — soft decision tree with entmax routing; differentiable but soft. **SYMPOL** — symbolic tree directly optimised as the policy, recovering hard, inspectable decisions. All baselines use PPO with GAE; hyper-parameters come from the original Optuna-tuned `configs.py`.

Q0 — Baseline Reproduction (PPO, 11 environments)

Five seeds $\times 10^6$ steps per method–environment pair. Mean episodic return $\pm$ std; bold = higher mean.

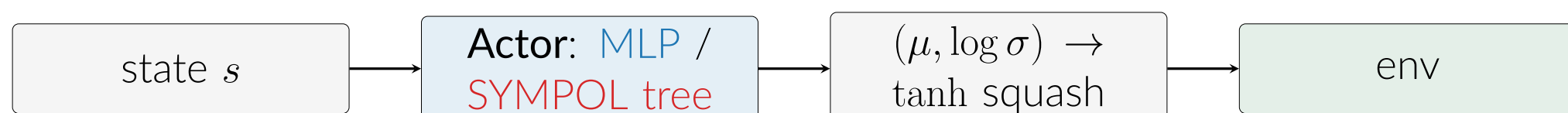
Domain	Environment	MLP	SDT
Classic Control	CartPole-v1	500.00 $\pm$ 0.00	500.00 $\pm$ 0.00
	Acrobot-v1	−74.36 $\pm$ 6.16	−76.28 $\pm$ 3.85
	Pendulum-v1	−163.40 $\pm$ 24.23	−292.78 $\pm$ 159.45
	MountainCarCont.-v0	75.36 $\pm$ 37.93	−0.00 $\pm$ 0.00
Box2D	LunarLander-v2	145.91 $\pm$ 127.91	−155.30 $\pm$ 47.66
MiniGrid	Empty-Random-6x6	0.844 $\pm$ 0.226	0.959 $\pm$ 0.006
	DoorKey-5x5	0.963 $\pm$ 0.003	0.959 $\pm$ 0.006
	LavaGapS5	0.948 $\pm$ 0.001	0.911 $\pm$ 0.074
	LavaGapS7	0.837 $\pm$ 0.093	0.837 $\pm$ 0.093
	DistShift1	0.949 $\pm$ 0.005	0.939 $\pm$ 0.000
	DistShift4	0.815 $\pm$ 0.142	0.960 $\pm$ 0.013

Takeaway

- **Discrete actions:** MLP and tree-based policies are competitive (differences within seed noise).
- **Continuous control:** the MLP clearly wins; soft routing cannot form the sharply-peaked Gaussians needed for precise actions (SDT *fails* on MountainCar and LunarLander).
- This continuous-control weakness is exactly what motivates studying SYMPOL off-policy (E5) and its robustness (E6).

E5 — SYMPOL Meets SAC (off-policy)

We re-implement SAC in JAX/Flax and **swap only the actor** (twin critic, Polyak target, auto-temperature, replay buffer all shared). **Plan A: hard routing + single-leaf Gaussian** — the tree collapses to one root-to-leaf path via a straight-through estimator, giving a **closed-form** $\log \pi$ for SAC’s entropy term.



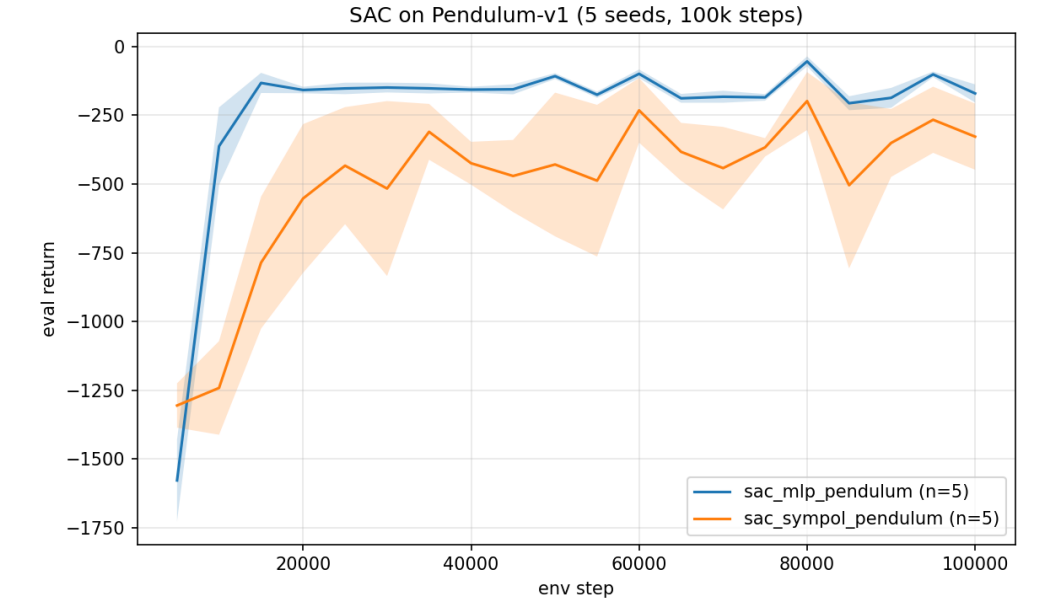
The actor is the *only* changed component; both actors output $(\mu, \log \sigma)$ of identical shape, so the squash and log-prob correction are shared. **Results** → top of right column.

E5 — Results [Feasibility: YES]

Actor	Final @100k	Best eval
SAC + MLP	−170.3 $\pm$ 32.8	−52.9 $\pm$ 17.2
SAC + SYMPOL	−327.5 $\pm$ 120.3	−130.4 $\pm$ 34.2

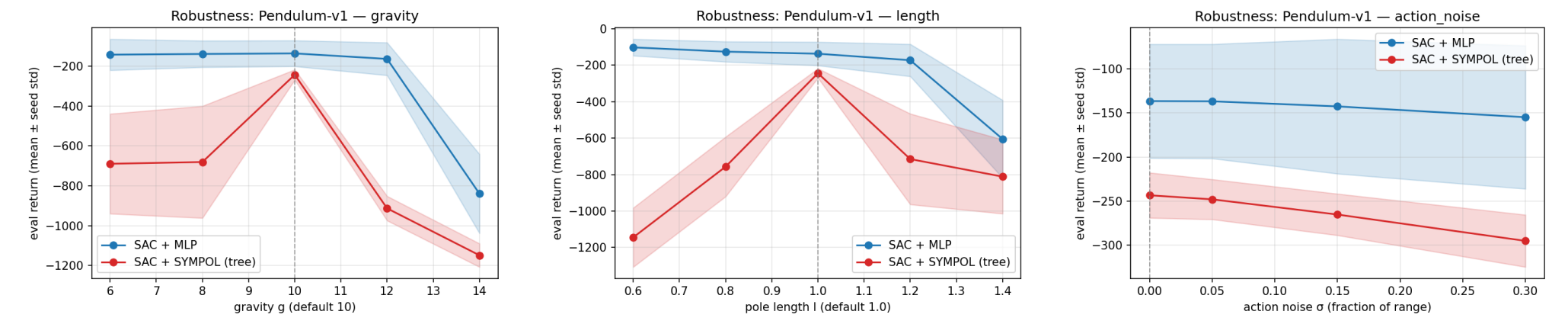
Pendulum-v1, depth-7 tree (128 leaves), 5 seeds $\times$ 100k steps, untuned CleanRL settings.

Result: the tree *does* learn off-policy (random $-1350 \rightarrow$ working controller) — feasibility confirmed — but converges **slower** and with **higher seed variance** than the MLP.



E6 — Robustness Under Perturbation [Hypothesis: REJECTED]

Policies trained *only at nominal physics* are evaluated **zero-shot** across perturbed environments. Dashed line = training value; bands = seed/episode spread.



- **Physics shift (gravity, pole length):** the **MLP** holds a wide $\pm 20\text{--}40\%$ plateau; the **depth-7 tree** is *sharply peaked* at the nominal dynamics and degrades steeply off-nominal (e.g. $l = 0.6 \rightarrow -1145$).
- **Actuator noise:** both degrade *gently* (at $\sigma = 30\%$: MLP -13% , tree -21%).
- **So the “trees are inherently robust” intuition fails under physics shift.** A large tree carves fine, nominal-specific thresholds; a shifted state crosses into a cell with the *wrong* constant action, and hard routing offers no smoothing.

Conclusions & Future Work

- Baselines reproduce the SYMPOL envelope: trees rival MLPs on discrete tasks, lag on continuous control.
- **E5:** an interpretable tree *can* be trained off-policy under SAC — a new, working extension.
- **E6:** the assumed robustness advantage does *not* hold under physics shift; the tree overfits the training dynamics. **Prediction:** a *shallower* tree should be more robust.
- **Next:** depth ablation, soft-routing leaf mixtures, and CartPole / MountainCar perturbation sweeps.

References

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